

Computer Science & IT

Database Management System



**File organization
And
Indexing**

Lecture No. 01



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Recap of Previous Lecture



✓ Topic

Basic time stamp ordering protocol

✓ Topic

Time stamp ordering protocol with Thomas write rule

✗ Topic

Wait-die protocol for deadlock prevention

✗ Topic

Wound-wait protocol for deadlock prevention

Topics to be Covered



Topic

Database, File and Records



Topic

Organization of Records



Topic

Categories of index





Topic : Database-File-Records

- Database is collection of files
 - * Files are used to store records
 - * Disk blocks are used to store the records of a file
- Records of different files
Can not be stored in
the same block of disk

Database

Student

Course

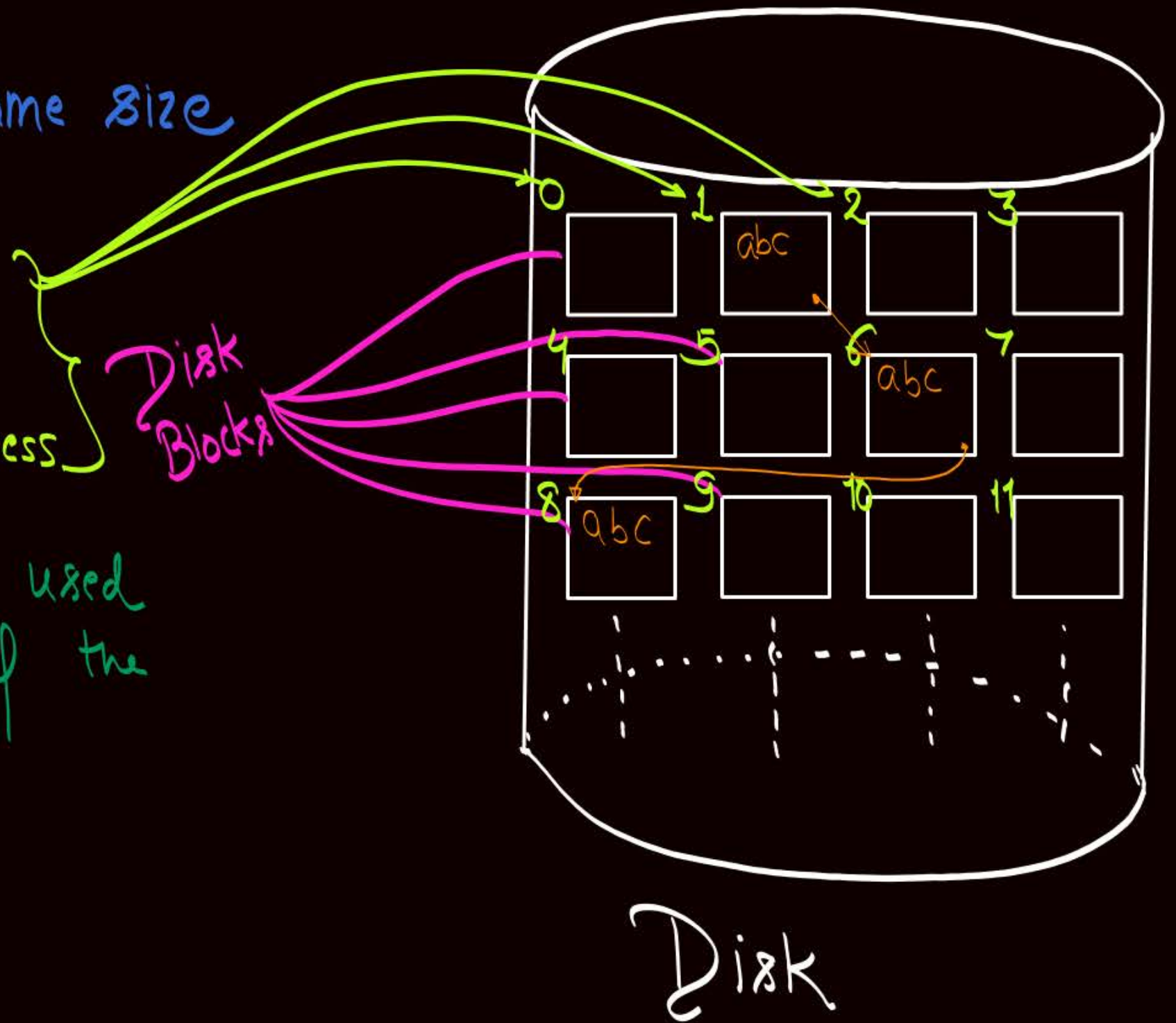
Enroll



* Each disk block within the disk will be of same size

* Each block of disk is identified by a unique no. known as disk block address

* Each disk block can be used to store multiple records of the same file





Topic : Types of Records

There are two types of records

(1) Fixed length records

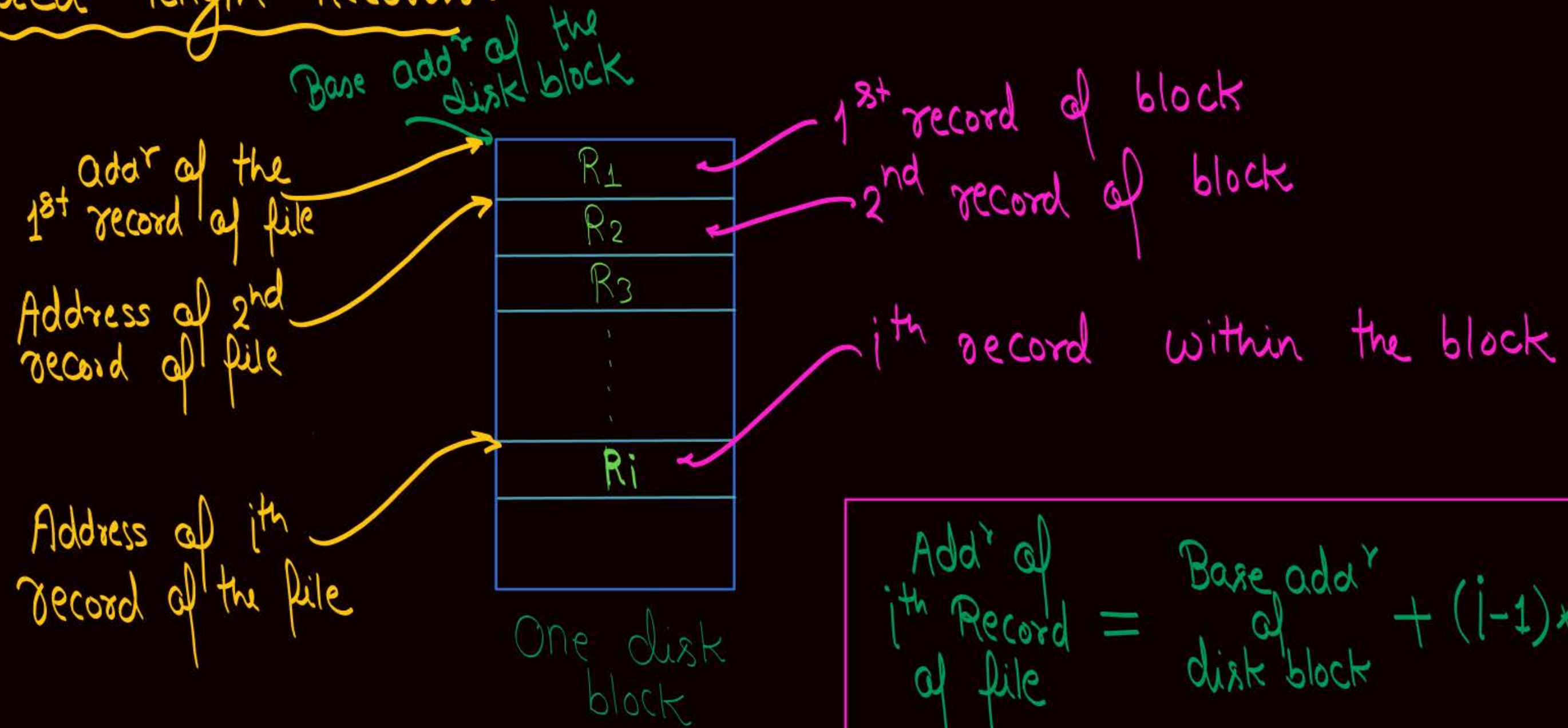
(2) Variable length records



Topic : Fixed length records

→ When each record of the file is of same size

Fixed length Records: -



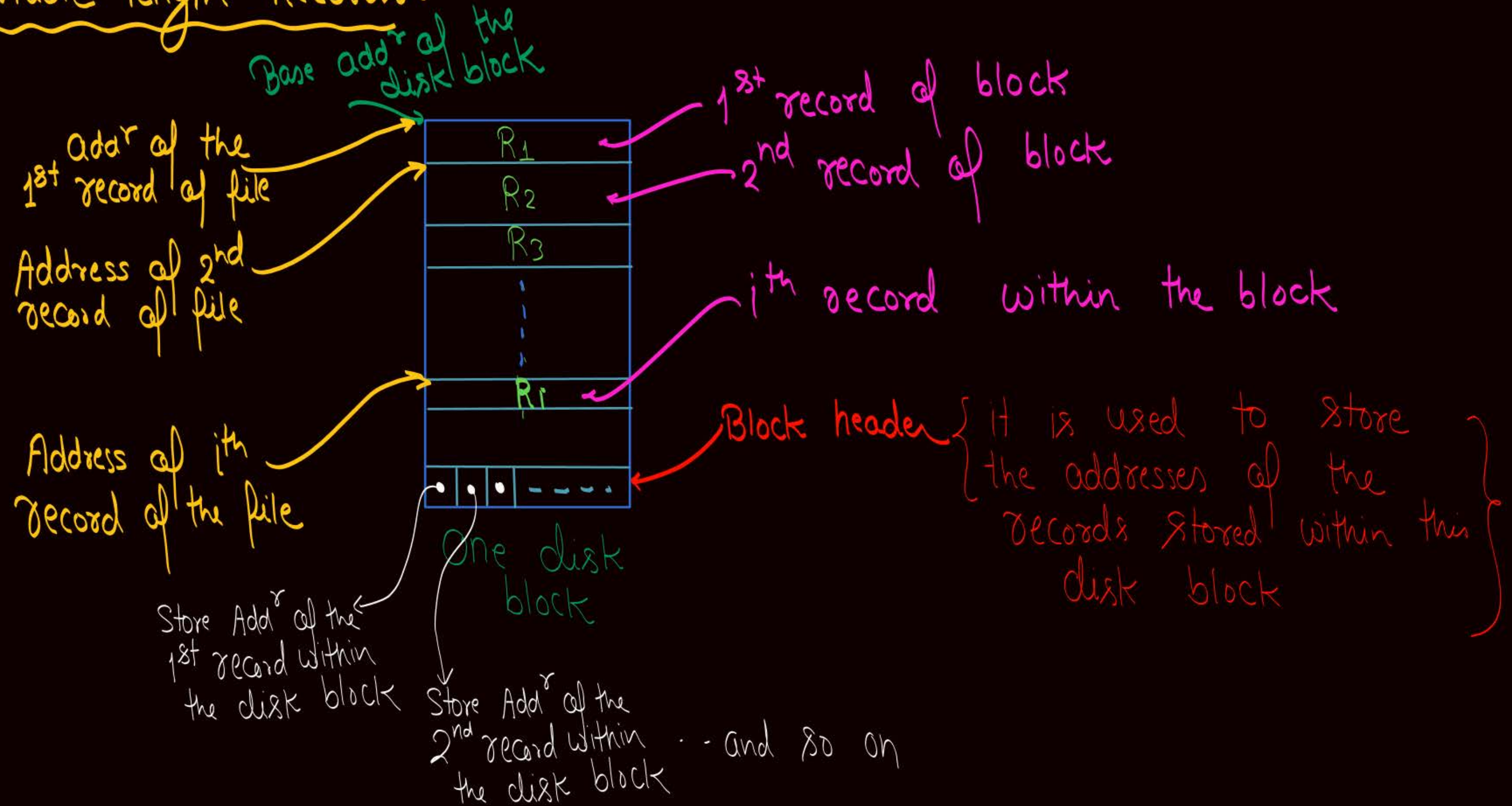
$$\text{Addr of } i^{\text{th}} \text{ Record of file} = \text{Base addr of disk block} + (i-1) * \text{Size of each record}$$



Topic : Variable length records

→ When records of the same file may be of different sizes.

Variable length Records: -



Note →

Block header may be required with "Fixed length records" as well in order to store the add^r of next disk block used to store the

Consecutive records of the file if it will be required if "link allocation" is used to allocate the blocks of the disk }

→ The pointer used to store address of the next disk block of the file is called "Block Anchor".

Note :-

- ① If nothing is given in the question about Block header size, then it is considered "Zero".
i.e, the complete disk block size will be used to store the records of the file
- ② If Block header size is given in the question, then effective space available to store the records of the file will be :- $(\text{Disk block size} - \text{Block header size})$



Topic : Blocking Factor



Blocking factor (B_f) of a disk block w.r.t. database file is defined as average number of records stored per block of the disk.

$$\text{Blocking factor } (B_f) = \frac{\text{Disk block size}}{\text{Avg. record size}} \quad \left\{ \begin{array}{l} \text{When Block header size} \\ \text{is not given} \end{array} \right\}$$

$$\text{Blocking factor } (B_f) = \frac{(\text{Disk Block size} - \text{Block header size})}{\text{Avg record size}} \quad \left\{ \begin{array}{l} \text{When Block} \\ \text{header size} \\ \text{is given} \end{array} \right\}$$



Topic : IO cost



IO cost of a record access can be defined as number of disk blocks that needs to be transferred from secondary memory to main memory in order to access that record



Topic : Organization of Records

There are two ways in which records of the file can be organized within the disk block

- ① Unspanned organization
- ② Spanned organization



Topic : Un-spanned Organization

Consider,

Disk block size = 100 Bytes

Record size = 40 Bytes

$$\text{Blocking factor (Bf)} = \left\lfloor \frac{\text{Block size} - \text{Block header size}}{\text{Record size}} \right\rfloor$$

w.r.t. Unspanned-organization

$$\text{Blocking factor (Bf)} = \left\lfloor \frac{100 - 0}{40} \right\rfloor = 2$$

B ₁	R ₁ (40 Bytes)
	R ₂ (40 Bytes)
	/// 20 Bytes ///
B ₂	R ₃ (40 Bytes)
	R ₄ (40 Bytes)
	/// 20 Bytes ///
	⋮

ie, Complete record must be stored within the same block

Available space is not sufficient to store the record completely
∴ this space is not used

Hence Unspanned organization can result in internal fragmentation



Topic : spanned Organization

A single record may be allowed to span in more than one block

Consider,

Disk block size = 100 Bytes

Record size = 40 Bytes



$$\text{Blocking factor (Bf)} = \frac{\text{Block size} - \text{Block header size}}{\text{Record size}}$$

w.r.t. Spanned organization

$$\text{Blocking factor (Bf)} = \frac{100 - 0}{40} = 2.5$$

Using spanned orgⁿ there is no internal fragmentation
{ Except for the last block of the file }



Topic : Search-key & Type of File

Need not have
Unique value



Search key :- Attribute/field used to search for a record
in a database file is called search Key

Employee database File

Retrieve Record of
the Employee
with $Eid = E_6$

Search for record
will be performed
based on "Eid"

∴ Eid will be the
Search Key

Eid	--other attributes--	Dept-id
E_3		2
E_1		1
E_4		2
E_6		3
E_2		4
E_5		1
⋮	⋮	⋮

$\left. \begin{array}{l} B_1 \\ B_2 \\ B_3 \end{array} \right\}$

Retrieve records of
all the employees
w.r.t. $dept_id = 2$

Search will be
performed based on
"dept-id"

∴ Dept-id will be the
Search Key



Topic : Search-key & Type of File

Need not have
unique value



Search key :- Attribute/field used to search for a record in a database file is called search key

Types of file :- There are two types of files

(1) Unordered file :- If records of the file are not Physically ordered (Unsorted) based on Search Key, then file is Unordered file

(2) Ordered file :- If records of the file are physically Ordered (sorted) based on Search Key, then file is Ordered file

Retrieve Record of the Employee with Eid = E6

Search for record will be performed based on "Eid"

∴ Eid will be the Search Key

Records of the file are not Ordered based on "Eid"
∴ Unordered file

Employee database file

Eid	--other attributes--	Dept-id
E3		2
E1		1
E4		2
E6		3
E2		4
E5		1
⋮	⋮	⋮

$\left. \begin{array}{l} \text{B}_1 \\ \text{B}_2 \\ \text{B}_3 \end{array} \right\}$

Retrieve records of all the employees w.r.t. dept-id = 2

Search will be performed based on "dept-id"

∴ Dept-id will be the Search Key

Records of the file are "Not ordered based on Dept-id"
∴ Unordered file

Employee database file

Retrieve Record of
the Employee of
Eid = E6

Search will be performed
based on Eid.
∴ Eid is the Search
Key

Record of the file are
Physically ordered based
on Search Key i.e. Eid.
∴ File is ordered file w.r.t.
given situation

Eid	--other attributes--	Dept-id
E1		1
E2		4
E3		2
E4		2
E5		1
E6		3
:	:	:

Records are still
Unordered w.r.t. "dept-id"

∴ If Search Key
is the dept-id,
then file will be
Unordered file



Topic : IO cost without indexing

→ Let "N" is the total number of disk blocks required to store the given database file,

(i) Worst case IO cost = N {search will be linear search}
(When file is un-ordered)

(ii) Worst Case IO Cost = $\lceil \log_2 N \rceil$ {If file is ordered then}
(When file is ordered) {Binary search is possible}

* Note:- If database is too large, then "N" will also be a large value and in that case $\lceil \log_2 N \rceil$ will also be a significant value
∴ We must do something to reduce the IO Cost (Hence concept of indexing will be used)

Ordered file

Vs

Unorder file { Also known as Heap file }

① Binary search is possible

② Insertion of a new record in an ordered file will be a costly operation. (because record must be stored at appropriate position in file)

① Binary search is not possible

② Insertion of a new record will be easy. { We can simply insert the record in the last block of the file }



2 mins Summary



✓
Topic

Database, File and Records

✓
Topic

Organization of Records

Topic

Categories of index ✓

THANK - YOU